

# EXPERIMENT 17

## Edge Detection Using Sobel Matrix Along X-Axis

### Aim

To perform edge detection using the Sobel matrix along the X-axis on an input image using Python and OpenCV.

### Procedure

1. Import the required libraries, cv2 and numpy.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply the **Sobel operator along the X-axis** to detect vertical edges.
5. Convert the result into an appropriate display format.
6. Display the original image and the Sobel X edge-detected image.
7. Save the output image.

### Code

```
import cv2
import numpy as np

# Read the input image
img = cv2.imread("image.jpg")

# Check whether image is loaded
if img is None:
    print("Error: Image not found")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
# Apply Sobel operator along X-axis
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

# Convert to 8-bit image
sobel_x = cv2.convertScaleAbs(sobel_x)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Sobel X Edge Detection", sobel_x)

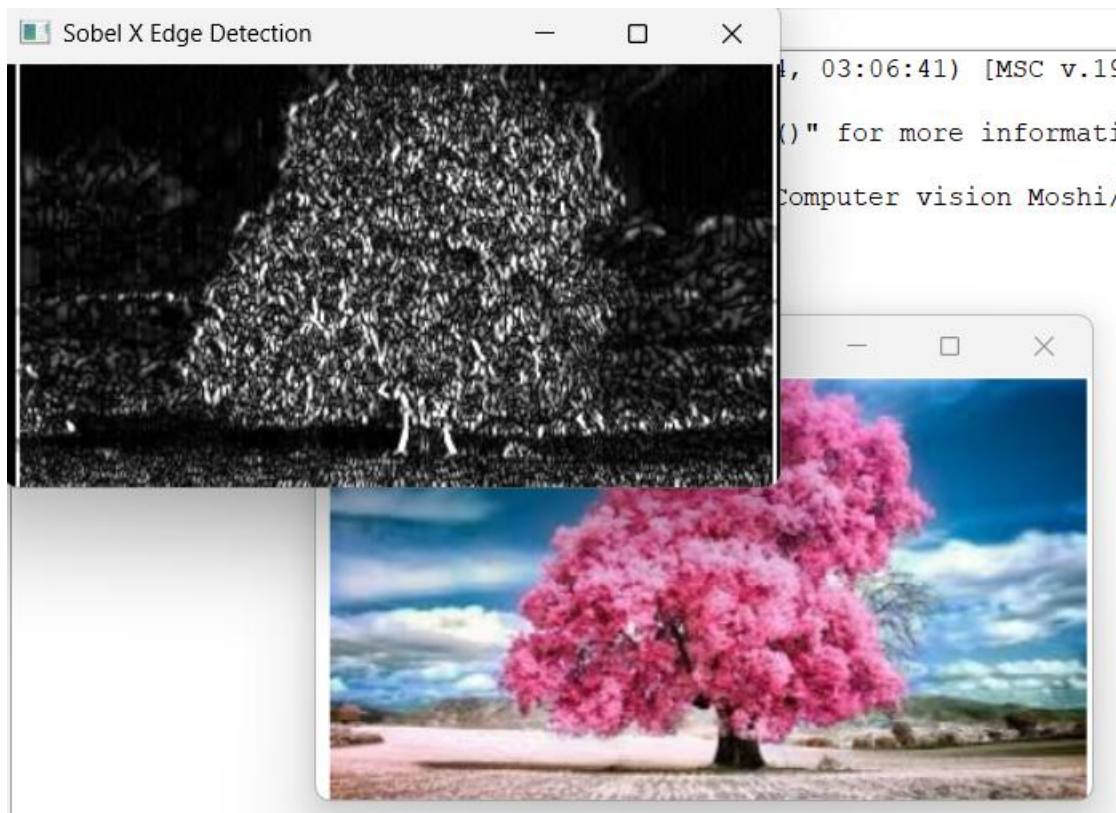
# Save output
cv2.imwrite("sobel_x_output.jpg", sobel_x)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

## Input



## Output



## Result

Thus, edge detection along the X-axis was successfully performed using the Sobel operator, and the vertical edges in the image were detected.

